TITLE OF THE PROJECT

(14 POINTS, BOLD, TIMES NEW ROMAN,1.5 LINE SPACE)

Report submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of

B. Tech. Write your branch of study here in black color

Submitted by

NAME (12 POINTS, BOLD, TIMES NEW ROMAN)
(Reg. No.: XXXXXXXXX)
NAME (12 POINTS, BOLD, TIMES NEW ROMAN)
(Reg. No.: XXXXXXXXX)

Month Year (Write in Black color)

SCHOOL OF ELECTRICAL & ELECTRONICS NGINEERING

THANJAVUR, TAMIL NADU, INDIA – 613 401

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Signature of Project Supervisor :

Name with Affiliation :

Date :

Project *Viva voc*e held on _____

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Declaration

I/We declare that the thesis titled ‟**Project Title (Title Case – First Letter Caps)**‟ submitted by me/us is an original work done by me/us under the guidance of **Name of the Guide, Designation**, **School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the Sixth semester of the academic year **2025-26**, in the **School of Electrical and Electronics Engineering**. The work is original and wherever I/we have used materials from other sources, I/we have given due credit and cited them in the text of the thesis. This thesis has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidate(s) :

Name of the candidate(s) :

Date :

Acknowledgements

Note: **Discuss with your guide and prepare the acknowledgement**

Acknowledgement

We would like to express our sincere gratitude to our Honourable Chancellor, **Prof. R. Sethuraman**, for providing us with the opportunity and the necessary infrastructure to carry out this project as part of our curriculum.

We extend our heartfelt thanks to our Honourable Vice-Chancellor, **Dr. S. Vaidhyasubramaniam**, and **Dr. S. Swaminathan**, Dean, Planning & Development, for their constant encouragement and strategic support throughout our academic journey.

We would also like to thank **Dr. R. Chandramouli**, Registrar for providing us with the opportunity to undertake this project.

We express our sincere appreciation to **Dr. K. Thenmozhi**, Dean, SEEE; **Dr. R. John Bosco Balaguru**, Dean (Sponsored Research); and **Dr. A. Krishnamoorthy**, Associate Dean (Academics), for introducing and facilitating this project as part of our curriculum.

We would like to convey our deep sense of gratitude to our **project guide**, Professor, School of Electrical & Electronics, for …….…….
We sincerely thank the P**roject Review Panel Members** for their valuable suggestions and constructive feedback throughout the course of this project. Their insightful observations greatly contributed to improving the quality and depth of our work.

We extend our appreciation to the all the teaching and non-teaching staff of the School of Electrical & Electronics for their support and assistance, directly or indirectly, in the completion of this project.

Finally, we gratefully acknowledge the encouragement and support of our family and friends, which played a vital role in the successful completion of this project.

GUIDELINES TO PREPARE PROJECT REPORT

PARTS OF PROJECT REPORT

S. No.	Item Description	Page Number
1.	Detailed Instructions to prepare the chapters of the Report	Refer Page 8 to 16 of this document
2.	Cover page	Refer Page 17
3.	Inside cover page	Same as cover Page
4.	Bonafide Certificate	Refer Page 18
5.	Declaration by Student	Refer Page 19
6.	Certificate (if any, from Industries or other organisations)	**You have to attach the certificate issued by the Industry or Institute or other organisations**
7.	Acknowledgements (Not mandatory)	Refer Page 20
8.	Abstract	Refer Page 21 Maximum of 300 words and should contain no more than six key words.
9.	Table of contents	Refer Page 22
10.	List of Figures	Refer Page 23
11.	List of Tables	Refer Page 24
12.	Abbreviations (if any)	Refer Page 15
13.	Notation (if any)	Refer Page 16
14.	Chapters	Refer Page 25
15.	Conclusion	Refer Page 28
16.	References	Refer Page 29
17.	Appendices	**Consult with you guide and include appendix accordingly.**
*** Number of report copies to be submitted (soft bound)**		**One per project + 1 Soft copy**

STYLE NOTES FOR PROJECT REPORT PREPARATION

Use A4 (210 mm x 297 mm) bond un-ruled paper (80 GSM) for all copies submitted. Use one side of the page for all printed/typed matter. It should be hard bound with **WHITE** title page for final semester project. For mini project, do soft binding with cover page on photo finish paper

Numbering

Pages

Every page in the Project report, except the title page, must be accounted for the page numbering, starting from acknowledgements and till the beginning of the introductory chapter, should be printed in small **Roman numbers, i.e., i, ii, iii, iv, etc.**,

All printed page numbers should be located at the bottom centre of the page, 17 mm (2/3”) from the bottom edge, using normal print.

Chapter

Use only Arabic numerals. Chapter numbering should be centre on the top of the page using large bold print.

Example: **CHAPTER 1**

Sections

Use only Arabic numerals with decimals. Section numbering should be left justified using bold print.

Example:
1.1, 1.2, 1.3, etc.

Equation(s)/ Formula(e)

Use only **Arabic numerals** with single decimal. Equation numbers should be right justified using normal print. Mathematical symbols should be printed in *italics.*

Format: (<Chapter number>.<Equation serial number)

Example:

$$\mathcal{E} = \left[\frac{2\cot\left(\phi + \frac{\psi}{2}\right) + \psi \cos ec\left(\phi + \frac{\psi}{2}\right)}{\sqrt{3}} \right] \tag{1.1}$$

$$\int_0^{\bar{\varepsilon}_f} \sigma_\theta d\bar{\varepsilon} = C_1 \tag{1.2}$$

$$I_1(\bar{\varepsilon}_f) = \frac{2}{3}(2\alpha+1)\int_0^{\bar{\varepsilon}_f} \bar{\sigma} d\varepsilon_z = \frac{2}{3}(2\alpha+1)\frac{k(\bar{\varepsilon}_f)^{n+1}}{n+1} \tag{1.3}$$

Please note that the equation numbers are flush right in normal print.

TEXT

Colour: Black print
Font:

Regular text	- Times New Roman 12 pts., and normal print.
CHAPTER HEADINGS	Time New Roman 14 pts. And **bold** print and all capitals.
SECTION HEADINGS	Time New Roman 12 pts. And **bold** print and all capitals.
Subsection Headings	Time New Roman 12 pts. And **bold** print and leading capitals, i.e., only first letter in each word to be the capital.
Special Text	Italics/Superscript/Subscript/Special symbol, etc, as per necessity. Special text may include footnotes, endnotes, physical or chemical symbols, mathematical notations, etc.
References	Same font as regular text. Serial number and all authors names to be in bold print. Journal names and book titles in italics. For format see Annexure 9.

Spacing:

Use **1. 5 spacing** between the lines.
Use **a single blank line** between paragraphs.

All paragraphs in the project report should be justified from the first line the last line.

Use **double spacing** between the regular text and quotations.

Provide **one blank line** between:

(a) Chapter title and first sentence of a Chapter.

Use single line spacing:

(a) In footnotes and endnotes for text,
(b) In explanatory notes for tables and figures.
(c) In text corresponding to bullets, listings, and quotations in the main body of the thesis.

Use **single space** in references and **double space** between references.

Justification: The text should be **fully justified.**

Hyphenation should be avoided as far as possible.

Text corresponding to bullets and listings should be indented.

Quotations from other research work must be indented on the left and the right if they are longer than two lines. Shorter quotations can be included as a part of the regular text.

Windows & Orphans: At the bottom of a page, a paragraph should have atleast two lines. Similarly, at the top of a page, a paragraph should end with atleast two lines.

MARGINS

The margins for the regular text are as follows:

LEFT	= 1.25”
RIGHT	= 1”
TOP	= 1”
BOTTOM	= 1”

Please note that the bottom of the page numbers should be 17 mm above the bottom edge of the numbered pages.

TABLES

A sample for tables is provided in page 26

All tables should have sharp lines, drawn in black to separate rows/columns as and when necessary.

Tables should follow immediately after they are referred to for the first time in the text. Splitting of paragraphs, to include tables on a page, should be avoided. Provide **three spaces** on the top and the bottom of all tables to separate them from the regular text, wherever applicable.

The last line of the title of any table should be 10 mm to 15 mm **above the top-most horizontal line of the table**, and the title should be centered concerning the table. The titles must be in the same font as the regular text and should be single-spaced. The title format is given below:

Table<blank><chapter number>.<serial number><left indent><table title>.

Example (of a small table which is sought to be placed within the text):

Table 1.1 Results of the simulation experiment.

The contents of the table will be within the surrounding double line (which indicates the top-most, left-most, right-most, and bottom-most boundaries of the table)

Whenever a table exceeds one-page present the full title of the table on the first page and on the following pages, provide the table number and state "(cond.)" after it.

Example: (notice the left justification)

Table 1.2 (contd.)

Wherever explanatory notes are used for clarifying any information presented in the tables, print them after leaving a single space immediately below the tables.

All tables in landscape format must be placed such that their top portions are near the binding of the thesis and their bottom portions near the outer edge.

FIGURES

Sample figure is shown in page 27

All figures, drawings, and graphs should be drawn in black with sharp lines and adequate contrast between different plots if more than one plot is present in the same graph.

Figures should follow immediately after they are referred to for the first time in the text. Splitting of paragraphs, to include figures on a page, should be avoided. Provide three spaces on the top and the bottom of all figures to separate them from the regular text, wherever applicable.

The first line of the title for figures, drawings, graphs and photos should be between 10 mm and 15 mm below the bottom, and they should be centred concerning the figure. The titles must be in the same font as the regular text and should be single-spaced. The title format is given below:

Fig.<blank><chapter number><serial number><left indent><figure title>.

Example:

Fig. 1.1. Stability of the RF device.

Wherever a figure exceeds one page (as in the case of large flow charts for computer programmes) present the full title of the figure on the first page and in the following pages provide the figure number and state "(contd.)" after it.

Example: (notice the left justification)

Fig.1.1**(contd.)**

When there are many plots in a single graph or figure, the lettering, labelling or numbering of each plot for its identification should be of a size such that even after size reduction in the report, the identification should be legible.

All Figures in landscape format must be placed such that their top portions are near the binding of the report and their bottom portions near the outer edge.

PHOTOS

Use Colour photos only if they are necessary. Remember that the report may have to be photocopied. In case Colour photos are used, all copies of the report must contain only colour photos.

Photos may be printed on glossy paper, and be mounted with white case in (e.g., Elmer's glue), glue stick, dry mounting tissue, or any good adhesive. Do not use rubber cement or cello tape.

Each photo should be numbered and referred to as a figure. Photo titles should be similar to those provided for figures.

DRAWINGS

Drawings which are larger than A4 size are not encouraged. If larger drawings are absolutely necessary they may be suitably folded to A4 size in the report.

Each drawing should be numbered and referred to as a figure. Drawing titles should be similar to those provided for figures.

FOOTNOTES

In presenting footnotes and references, use a consistent form acceptable in your discipline.

PUNCTUATION

Please refer to any standard style manual such as the Chicago *Manual of Style*[1], where rules of punctuation are clarified. Student must note that different styles are in practice these days.

For example, some people insert a full stop before ending a sentence in double quotes, whereas other inserts the full stop after the double quotes. Both styles are in practice.

STATISTICAL ANALYSIS

[1]Chicago *Manual of Style, Prentice Hall of India, New Delhi, 1989*

Statistical methods used to analyse the data should be clearly mentioned in the methodology section. Meaning of symbols used in the graphs should be clearly explained in the legends for figures and graphs.

APPENDICES

If the project requires clearances from competent authorities such as Institutional ethics committee (IEC), Institutional Biosafety Committee (IBC), etc., a copy of the approval letter/s should be appended in the appendix.

REFERENCES

References should be arranged in alphabetical order of surname or last name of the first author.

Two or more references by same author(s) in the same year should be indicated by small case alphabets in italics.

For papers by one author:

Bensamaia SJ (2015) Biological and bionic hand: natural neural coding and artificial perception. *Philosophical Transactions of the Royal Society London B Biological Sciences*, **19**, 370-375.

For papers by two authors:

Ghafoor U, Kim S and Hong KS (2017) Selectivity and longevity of peripheral nerve and machine interfaces: A review. *Frontiers in Neurorobotics,* **11,** 59-63**.**

For papers by more than two authors:

Knutson JS, Fu MJ, Sheffler LR and Chae J (2015) Neuromuscular electrical stimulation for motor restoration in hemiplegia. *Physical Medicine and Rehabilitation Clinics of North America, **26 (4)**, 729-745.*

Adewole DO, Serruya MD, Harris TP, Burrell JC, Petrov D, Chen HI, Wolf JA and Cullen DK (2016) The evolution of neuroprosthetic interfaces. *Critical Reviews in Biomedical Engineering, **44**, 123-152.*

Note: Indicate the names of all authors, if feasible, and avoid the use of *et al* (to indicate the presence of two or more co-authors) after the first author's name.

For books:

Sanchez JC(2016) Neuroprosthetics: Principles and Applications. *CRC Press (USA).*

For papers presented at Conferences:

Young. E.F. and H.L. Miller (2009) "Old biotechnology to new bio-technology: Continuum or disjunction?" Proceedings of *International Biosymposium for Advancement of Biotechnology*, Tokyo, October, 6-85.

SIMILARITY CHECK REPORT

- A report on similarity check obtained from Turnitin® shall be submitted along with the project report/ thesis.
- No more than three words in a sentence should be taken from a previous report or publication. Use quotation mark and cite the reference where words and sentences are taken from another work.
- If a figure, table or any data is reproduced from a previous study, prior permission from competent authority should be obtained. Make a mention of the authority from which such permission was obtained.
- References should be cited appropriately when information from previous studies are mentioned in the project report.

ABBREVIATIONS

The student must take utmost care in the use of technical abbreviations. For example, “KM” stands for “Kelvin Mega” and not kilometre (which should be abbreviated as km) and “gms” stands for “gram meter second” and “grams” (which should be abbreviated as g). In addition, abbreviations pertinent to any specific discipline should be listed in alphabetical order as shown below.

MSA	Microstrip Antenna
PA	Patch Antenna
VSWR	Voltage Standing Wave Ratio
RP	Radiation pattern
LP	Linear Polarization
CP	Circular Polarization

NOTATIONS

The student must explain the meaning of special symbols and notations used in the thesis. Define English symbols, Greek symbols, and Miscellaneous symbols separately. Some examples are presented below.

English Symbols (in alphabetical order)

S	Scattering Parameters
E	Electric Field Intensity
H	Magnetic Field Intensity
Q	Charge
v	Velocity m
Meter	

Greek Symbols (in alphabetical order)

α	Rate
$\sigma(x)$	the standard deviation of x
$\sum$	Summation
Ω	Ohm
∞	Infinity

Miscellaneous Symbols (in alphabetical order)

$\lvert x \rvert$	absolute value of x

TITLE OF THE PROJECT

(14 POINTS, BOLD, TIMES NEW ROMAN,1.5 LINE SPACE)

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in partial fulfillment of the requirements
for the award of the degree of

B. Tech. Write your branch of study here in black color

Submitted by

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(Reg. No.: XXXXXXXXX)
NAME (12 POINTS, BOLD, TIMES NEW ROMAN)
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Signature of Project Supervisor :

Name with Affiliation :

Date :

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Examiner 1 **Examiner 2**

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Signature of the candidate(s) :

Name of the candidate(s) :

Date :

Acknowledgements

Note: Discuss with your guide and prepare the acknowledgement

Abstract

The body should be justified and typed in Times New Roman, 12 pt, not bold, with 1.5 line spacing and should be one page. The abstract should contain a brief background including the literature survey that leads to the research question/objective, methodology and the major results with overall conclusions.

For a team project, each member of the team shall write the abstract in his/her own words. Abstract shall also explicitly state in bullet format the specific contribution and specific learning by each individual. Project thesis / report shall have abstract of all the team members.

Specific Contribution

- **********
- **********
- **********

Specific Learning

- **********
- **********
- **********

Signature of the Guide | Student Reg. No :

Name : | Name :

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CHAPTER 1

INTRODUCTION

A micro strip patch antenna consists of a dielectric substrate, with a ground plane on the other Side. due to its advantages such as low weight, low profile configuration, low fabrication cost and capability to integrate with microwave integrated circuits technology, the micro strip patch antenna is very well suited for applications such as wireless communication system, GPS, cellular phones radar system and satellite communication system.

The development of small integrated antenna plays a significant role in the progress of rapidly expanding military and commercial communication applications. The technology to support these applications has been made possible by recent advance in high density RF and microwave circuit packaging. Assystem, requirements for faster data transmission in lighter compact design derive the technology area, higher frequency design solutions with large density layouts require integration of microwave devices, circuitry, and radiating elements that offer light weight, small size, and optimum performance. Over the past two decades, micro strip patch antenna has received considerable attention for use in personal communication systems applications due to its compactness among other advantages. Intensive research has been carried out to develop new techniques to overcome the micro strip patch antennas drawbacks, the most restrictive being narrow band. The bandwidth enhancement and its return loss improvement without increasing antenna size and production process is important to apply this antenna to the modern mobile communication systems.

Many applications in communications and radars require circular or dual linear polarizations, and the flexibility afforded by microstrip antenna technology has led to a wide variety of designs and techniques to fill this need. In recent years the demand for compact mobile Telephone handset has grown. Handsets with size of a pocket have begun appearing in the market, as the demand for increased electronic mobility grows, the need for small handsets will

most luckily increase. A small antenna size is required as one of the important factors in portable mobile communication systems.

The microstrip patch antennas (MPA) is widely being used because of its low volume and thin profile characteristics. The size of MPA is basically determined by its resonance length and width. The reduction of the patch size can be achieved by using patch substrate material with very high permittivity and small substrate height [2]. but; in this case, the low radiation efficiency will reduce the antenna gain.

In this study-shaped patch antenna optimized for simplicity in design and feeding is proposed .it has characteristic which will meet GPS system application. Parameters of the antenna such as return loss, impedance bandwidth, radiation patterns and gains are discussed. Extensive benefits of 3D view of radiation pattern facility provided by the ADS Software.

Table 1.1 Results of the simulation experiment

Parameter X	Parameter Y

A simple GPS repeater consists of an outdoor antenna, narrow band pass filters no noise amplifier and a reradiating indoor antenna. Figure1 show the black diagram of a simple GPS receiver. The outdoor antenna picks the GPS signal from the satellite. This antenna should be wide beam E-Shaped to acquire more satellites. The GPS received by the outdoor antenna is fed to input of narrow ceramic band pass filter. The filter signals then provided to a multistage low noise amplifier. The amplifier provides the necessary gain so that the signal can be retransmitted inside the building. With the help of indoor radiating antenna GPS signals is transmitted inside the buildings, so a wide beam and e shaped antenna is required micro strip antenna is the suitable for choice.

A GPS receiver can tell its own position by using the position data of itself, and compares that data with three are more GPS satellites. To get the distance to each satellite, by measuring the amount of time taken by radio signal (GPS signal) to travel from the satellite to the receiver

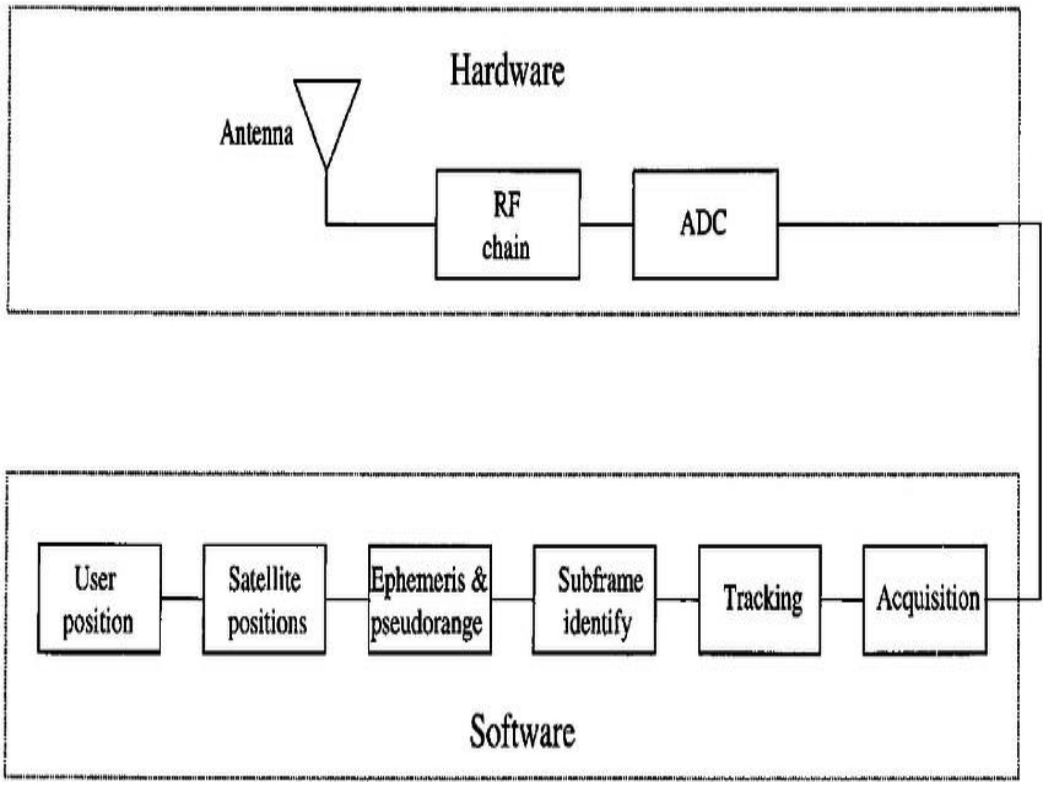

Fig.1 GPS receiver block diagram

Note: Ensure the diagrams drawn are of better quality. Do not scan and paste from the reference materials

The Global Positioning System (GPS) is a network of about 30 satellites orbiting the earth at an altitude of 20000 km... Once it has information on how far away at least 3 satellites are, your GPS receiver can pinpoint your location using a process called trilateration. PS satellites have atomic clocks on board to keep accurate time. The GPS receiver determines its position using three satellites to triangulate its 3D position.

Note: Discuss with your guide to formulate the number of chapters and Chapter Titles. Write the chapters as per the instructions given in the pages between 8 to 16

Conclusion

The body should be justified and typed in Times New Roman, 12 pt, not bold, with 1.5 line spacing and should be one page.

For a team project, ***each member of the team shall write the conclusion in his/her own words.*** It should include specific conclusion based on the understanding and work done by each individual.

Signature of the Guide Student Reg. No :

Name of the Guide: Name :

REFERENCES

1. **C.A. Balanis**, Antenna Theory, 2nd ed. New York: John Wiley & Sons, Inc., 1997.
2. **R. Bancroft,** Microstrip and Printed Antenna Design, Prentice Hall of India, 2006
3. **Wi S.H., Kim J.M., Yoo T.H., Lee H.J., Park J.Y., Yook J.G., Park H.K.** Bow-tieshaped meander slot antenna for 5 GHz application .in Proc. IEEE Int. Symp. Antenna and Propagation. 2, 456– 459, 2002
4. **J. R. James, P. S. Hall, and C. Wood,** Microstrip Antenna Theory and Design, Peter Perigrinus, London, 1981.
5. **B. L. Ooi, M.S. Leong, and Q. Shen,** "A novel equivalent circuit for E-shaped slot patch antenna," in Proc. IEEE Antennas and Propagation Society Int. Symp., 2001, vol. 4, pp. 482-485, 8-13 Jul. 2001.
6. **K. C. Gupta and A. B. Norwood**, Microstrip Antenna Design, Norwood, MA: Artech House, 1988.
7. **X. -X. Zhang and F. Yang,** "The study of slit cut on the microstrip antenna and its applications", Microwave Opt. Technol. Lett., vol. 18, no. 4, pp. 297-300, Jul. 1998.
8. **Shobhit K. Patel, Y.P.Kosta** "E-shape Microstrip Patch Antenna Design for GPS Application" Institute of Nirma University, 08-10 Dec, 2011, IEEE.

Annexure – I

Assessment scheme and the rubrics for award of marks

Review - 1: 30 Marks

I. Problem formulation/ Objective & Motivation/ Literature review or survey (Maximum marks: 10)	
	Marks
(a) Problem statement (5 marks)	
Clear definition and formulation of a problem statement	4-5
Fair definition and formulation of a problem statement	2-3
Poor/ill-defined problem statement	0-1
(b) Literature survey (5 marks)	
Exhaustive literature survey relevant to the problem	4-5
Satisfactory literature survey relevant to the problem	2-3
Inadequate literature survey	0-1
II. Methodology/Work plan (Maximum marks: 10)	
Detailed, well-defined, methodology/ work plan	8-10
Partial/incomplete methodology/ work plan	4-7
Ill-defined or unrealistic work plan	0-3
III. Work carried out (Maximum marks: 10)	
Clear evidence of work done in terms of carrying out analysis/fabrication/case studies/any other appropriate activity	8-10
Some work has been carried out	4-7
Very limited/no work has been carried out	0-3

Review - 2: 30 Marks

IV. Problem formulation/ Objective & Motivation/ Literature review or survey (Maximum marks: 10)	
	Marks
(a) Implementation of methodology (4 marks)	
Systematic implementation	4
Satisfactory implementation	2-3
Poor implementation	0-1
(b) Results (4 marks)	
Excellent presentation of data in terms of graphs, figures, tables & strict adherence to norms	4
Satisfactory presentation of data in terms of graphs, figures, tables & adherence to norms	2-3
Poor presentation of data & adherence to norms	0-1
(c) Adherence to codes, standards & ethical practices (2 marks)	
Full conformity	2
Partial conformity	1
Non-conformity	0
V. Interpretation/Understanding, Discussion & Conclusion(s) (Maximum marks: 20)	
(a) Interpretation (10 marks)	

In-depth analysis of data and interpretation of results	8-10
Satisfactory analysis of data	4-7
Insufficient/poor analysis	0-3
(b) Discussion (5 marks)	
Elaborate discussion with professional etiquette	4-5
Satisfactory discussion	2-3
Poor/insufficient discussion	0-1
(c) Attainment of Objectives/ Inferences/ Conclusions (5 marks)	
Successful attainment of objectives / meaningful inferences & conclusion(s)	4-5
Satisfactory attainment of objectives / reasonably meaningful inferences /conclusion	2-3
Non-attainment of objectives/no meaningful inference/conclusion	0-1

Guide's Evaluation: 20 Marks

VI. Project Report (Maximum marks: 10)	
	Marks
(a) Formatting (2 marks)	
Project report is prepared as per the format; references and citations are adequate	2
Major part of the report is as per the format with acceptable citations	1
Project report is not prepared according to format	0
(b) Technical content (5 marks)	
Report is well-written technically with excellent presentation of data in terms of graphs, figures, tables, discussion, inference/conclusion	4-5
Report is satisfactory with data, discussion and inference/conclusion	2-3
Report is unsatisfactory/poor	0-1
(c) Originality (3 marks)	
Similarity index is less than 10 %	3
Similarity index is between 10 % and 25 %	2 (Revision, if required)
Similarity index is between 25 % and 40 %	1 (Thesis requires re-submission)
Similarity index is greater than 40 %	0 (Thesis requires re-submission)

* While assessing the Similarity index, common phrases, known equations, nomenclature, generic terms and references are to be excluded.

VII(A). Assessment of Individual's Contribution to the Team (Maximum marks: 10)	
	Marks
(a) Synopsis (5 marks)	
Reflective report of the Synopsis correlates with the individual's contribution and learnings	4-5

Reflective report of the Synopsis partially correlates with the individual's contribution and learnings	2-3
Synopsis partially correlates /does not correlate with the individual's contribution and learnings	0-1
(b) Individual's contributions (5 marks)	
Individual's contribution in the execution of the project is evident through project diary/lab note, self and peer evaluation forms	4-5
Individual's contribution in the execution of the project is partly evident through project diary/ lab note, self and peer evaluation forms	2-3
Individual's contribution is not evident	0-1

***VII(B). Assessment of Individual's Learnings from the project (Maximum marks: 10)**	
	Marks
(a) Synopsis (5 marks)	
Reflective report of the Synopsis clearly describes the learnings	4-5
Reflective report of the Synopsis partly describes the learnings	2-3
Synopsis poorly describes/does not describe the learnings	0-1
(b) Individual's contributions (5 marks)	
Execution of the project is evident through project diary/ lab note	4-5
Execution of the project is partly evident through project diary/ lab note	2-3
Execution of the project is poor/unsatisfactory	0-1

* - for individual projects only

End Semester Evaluation: 50 Marks

I. Problem Formulation / Objective, Methods & Processes including experimental design, mathematical model and so on (Maximum Marks: 15)	
	Marks
(a) Problem statement/objective (3 marks)	
Clear definition and formulation of a problem statement/objective	4-5
Fair definition and formulation of a problem statement/objective	2-3
Poor/ill-defined problem statement/objective	0-1
(b) Literature survey (4 marks)	
Exhaustive literature survey relevant to the problem	4-5
Satisfactory literature survey relevant to the problem	2-3
Inadequate literature survey	0-1
(c) Methodology/Work plan (8 marks)	
Detailed, well-defined, methodology/ work plan	7-8
Partial/incomplete methodology/ work plan	3-6
Ill-defined or unrealistic work plan	0-2
II. Results and Discussion (Maximum marks: 15)	
(a) Quantum & Quality of Work (2 marks)	
Clear evidence of work done in terms of carrying out analysis/fabrication/case studies/any other appropriate activity	2
Some work has been carried out	1
Very limited/no work has been carried out	0

(b) Results (3 marks)	
Excellent presentation of data in terms of graphs, figures, tables & strict adherence to norms	3
Satisfactory presentation of data in terms of graphs, figures, tables & adherence to norms	1-2
Poor presentation of data & adherence to norms	0
(c) Adherence to codes, standards & ethical practices (2 marks)	
Full conformity	2
Partial conformity	1
Non-conformity	0
(d) Interpretation (6 marks)	
In-depth analysis of data and interpretation of results	5-6
Satisfactory analysis of data	2-4
Insufficient/poor analysis	0-1
(e) Discussion (2 marks)	
Elaborate discussion with professional etiquette	2
Satisfactory discussion	1
Poor/insufficient discussion	0
III. Project Report / Thesis (Maximum marks: 10)	
(a) Formatting (3 marks)	
Project report is prepared as per the format; references and citations are adequate	3
Major part of the report is as per the format with acceptable citations	1-2
Project report is not prepared according to format	0
(b) Technical content (7 marks)	
Report is well-written technically with excellent presentation of data in terms of graphs, figures, tables, discussion, inference/conclusion	6-7
Report is satisfactory with data, discussion and inference/conclusion	3-5
Report is unsatisfactory/poor	0-2
IV. Individual Contribution/Learning and Viva Voce (Maximum marks: 10)	
(a) Application of conceptual knowledge (5 marks)	
Application of knowledge is clearly evident	4-5
Limited application of knowledge	1-3
No evidence	0
(b) Viva Voce (5 marks)	
Answered most of the questions of the panel	4-5
Able to answer some of the key questions	2-3
Most of the questions not answered	0-1

Assessment Scheme for Review - I of Project

S.No	Name	Reg. No.	Problem formulation/ Objective & Motivation/ Literature review or survey (10)		Experimental Design / Mathematical Model / Solution Methodology / Methods & Procedures and so on (Methodology / Work plan) (10)	Work carried out (10)	Total (30)	Remarks
			Problem statement (5)	Literature survey (5)				
1								
2								

Assessment Scheme for Review - II of Project

S. No	Name	Reg. No.	Design/analysis/experiments/fabrication and following of ethical process in the same & Results (10)			Interpretation/Understanding, Discussion & Conclusion(s) (20)			Total (30)	Remarks
			Implementation of methodology (4)	Results (4)	Adherence to codes, standards & ethical practices (2)	Interpretation (5)	Discussion (5)	Attainment of Objectives/ Inferences/ Conclusions (5)		
1										

Project Supervisor's Mark (20)

S. No	Name	Reg. No.	Project Report (10)			Assessment of Individual's Contribution to the Team / Assessment of Individual's Learnings from the project (10)		Total (20)	Remarks
			Formatting (2)	Technical content (5)	Originality (3)	Synopsis (5)	Individual's contributions/ Execution of project (5)		
1									
2									

Assessment Scheme for End Semester Examination of Project

S.No	Name	Reg. No	I. Problem Formulation / Objective & Methods & Processes including experimental design, mathematical model and so on (15)			II. Results and Discussion (15)					III. Project Report / Thesis (10)		IV. Individual Contribution / Learning and Viva Voce (10)		Total (50)	Remarks
			I. (a)	I. (b)	I. (c)	II. (a)	II. (b)	II. (c)	II. (d)	II. (e)	III. (a)	III. (b)	IV. (a)	IV. (b)		
1																
2																

I. (a): Problem statement/objective (3)
I. (b): Literature survey (4)
I. (c): Methodology/Work plan (8)
II. (a): Quantum & Quality of Work (2)
II. (b): Results (3)
II. (c): Adherence to codes, standards & ethical practices (2)
II. (d): Interpretation (6)
II. (e): Discussion (2)
III. (a): Formatting (3)
III. (b): Technical content (7)
IV. (a): Application of conceptual knowledge (5)
IV. (b): Viva Voce (5)